

Investment Strategy Challenge 2026 · Proposal for Purple Magic Capital

The K-Dislocation Note

Turning the Iran-war oil shock into Korea's buy-the-dip trigger, with a maximum loss known on day one

KRW 135bn · 18-month note · 90% capital protected · Trade date 17 Sep 2026

Executive summary

One note, three catalyst engines, one known maximum loss

10%

Maximum loss

KRW ~13.5bn, known on day one: 90% capital protection

~14%

Option budget

Spent on three engines tied to oil, KRW and Korean equity catalysts

2 of 3

Regimes that pay

Pays in an oil shock and in a peace/re-rating; loses only in stagnation

We recommend **an 18-month, 90% capital-protected KRW note** on Brent, KOSPI 200 and USD/KRW to express our view that the Iran-war oil shock will dislocate Korean assets, and that structural inflows and reforms will drive the recovery. **Target redemption: 114-141% in oil-shock regimes, 109.5% in a ceasefire. Maximum loss: 10% (KRW 13.5bn $\approx$ 10 bp of AUM), contractual.**

Purple Magic needs asymmetry and trust

Client diagnosis: a sophisticated fund redeploying after a USD 30M loss

1

Who they are

- Korea-based hedge fund, USD 10bn AUM
- Macro and event-driven strategies
- Prefers customised, exotic structures

2

What they want

- Redeploy KRW 135bn (~USD 95–100M)
- Capture emerging macro catalysts
- Profit from dislocations and regulatory shifts

3

What they need

- A defined, known maximum loss
- Transparent pricing and Greeks
- Clear unwind terms and secondary market

Design brief: convex payoffs on Korea-specific catalysts, with a capped and disclosed downside.

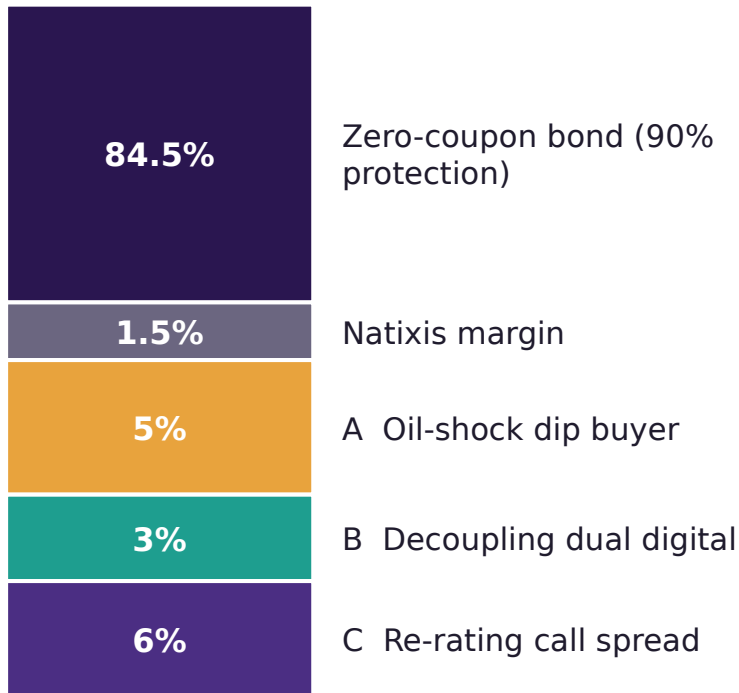
Five catalysts, one dislocation trade

Market view as of 17 Sep 2026 (status of each to be confirmed on Bloomberg)

1	Oil shock from the Iran war	Korea imports almost all its oil: trade balance, inflation and KRW are exposed	Engine A
2	AI memory cycle	Samsung and SK Hynix are a large share of KOSPI and exports	Engine C
3	Value-up reforms / MSCI watch	Governance reform and possible upgrade can re-rate Korean equities	Engine C
4	WGBI index inclusion	Index-driven buying of Korean government bonds supports KRW	Engine B
5	Fed vs BoK divergence	Drives USD/KRW and the Korea-US yield gap	Engine B

One note, three engines

How KRW 135bn is allocated inside an 18-month, 90% protected note



Oil-shock dip buyer

Brent $\geq 125\%$ (first 12M) switches on a KOSPI 200 call struck on the trigger day, ~120% participation



Decoupling dual digital

Pays ~9x if Brent $\geq 110\%$ AND USD/KRW $\leq 97\%$ at maturity



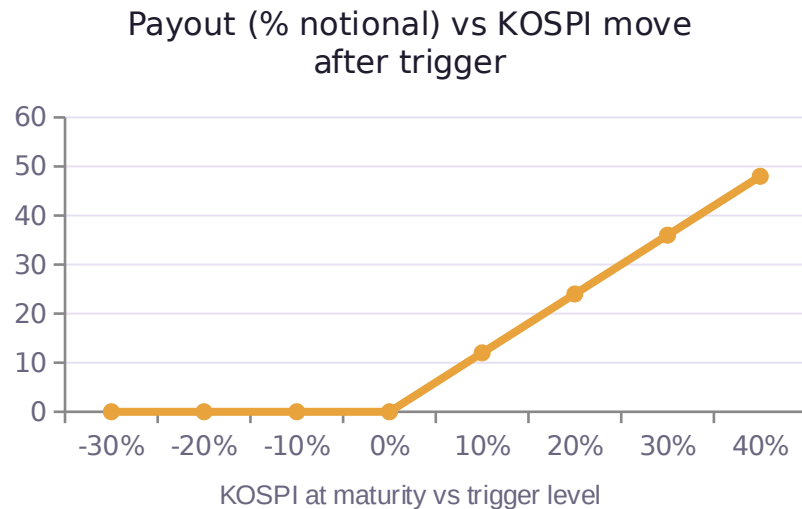
Re-rating call spread

~65% participation in KOSPI 200 from 100% to 130%

Engine A: the oil shock buys the dip

Brent-triggered KOSPI 200 call with strike reset

Trigger	Brent daily close $\geq$ 125% of initial, any day in months 1-12
On trigger	KOSPI 200 call starts, strike = KOSPI level on trigger day
Participation	~120% of KOSPI gain from trigger to maturity
Maturity	18 months
Budget	5% of notional
Cost vs vanilla	~4.1% per 100% participation, ~35-40% of an 18M ATM call

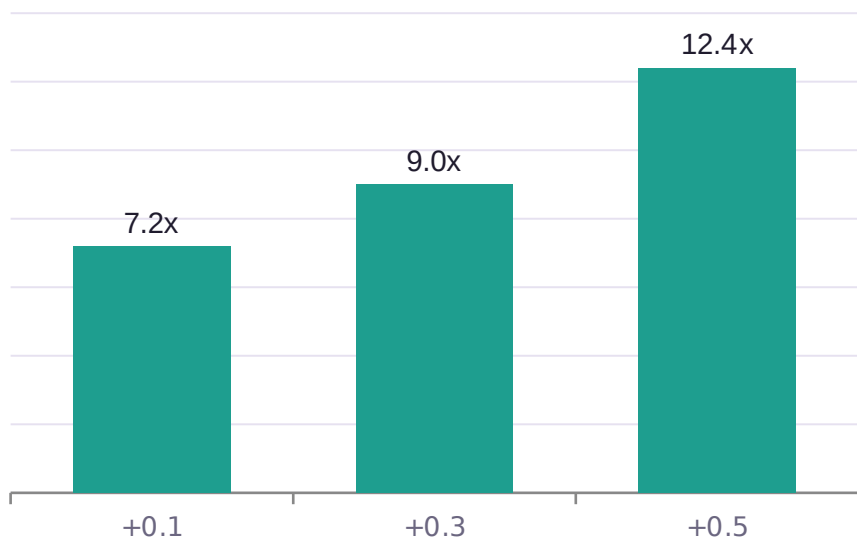


Why it's cheap: the call only exists if the shock happens (~44% trigger probability at 35% Brent vol), and its strike is reset to the dislocated level.

Engine B: KRW decouples from oil

Dual digital: Brent $\geq 110\%$ AND USD/KRW $\leq 97\%$ at maturity

Payout multiple vs corr(Brent, USD/KRW)



The view

WGBI bond inflows and chip exports keep KRW strong even if oil rises

Why it's cheap

Joint event goes against the usual oil-KRW link: the more positive the correlation, the cheaper the digital

Skew check

USD/KRW risk reversals favour USD calls, so the KRW-strength leg is the cheap wing (Unit 4)

Max loss

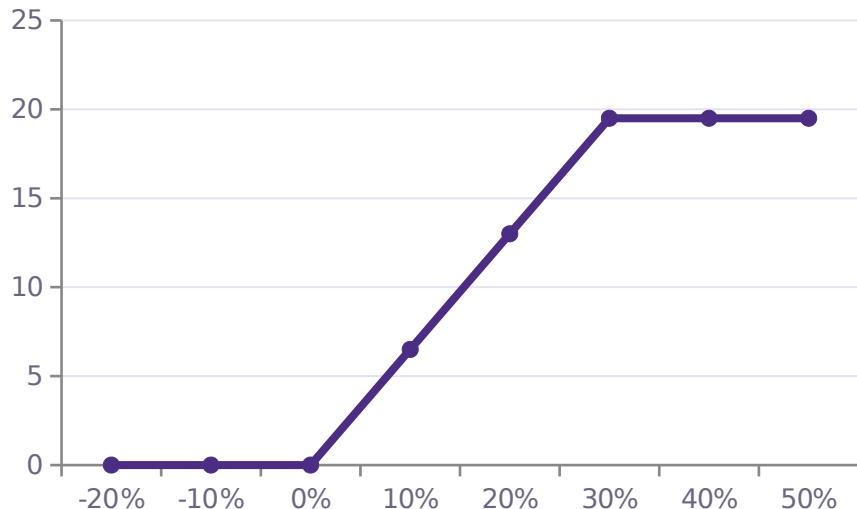
Premium only: 3% of notional

Illustrative prices from Monte Carlo with placeholder inputs; to be re-run with Bloomberg data as of 17 Sep 2026. Measure realised correlation (1Y, 3Y) before fixing strikes.

Engine C: owning the ceasefire

KOSPI 200 100/130 call spread, ~65% participation

Payout (% notional) vs KOSPI 200 return at maturity



Skew-aware pricing

9.2%

Cost with skew

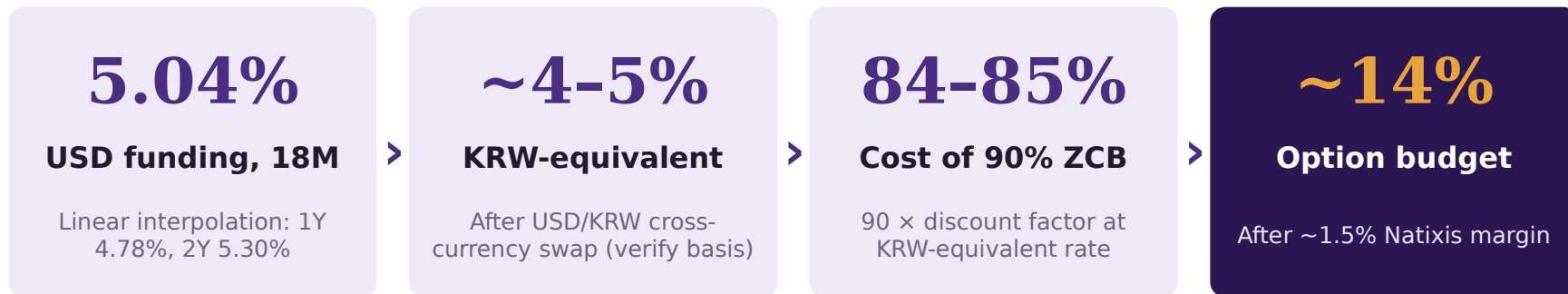
8.1%

Cost at flat vol

- The 130% call we sell sits at lower implied vol, so it earns less than flat vol suggests (Unit 4, slide 7).
- Budget 6% buys ~65% participation; maximum gain +19.5% of notional.
- Pays in the peace scenario, when Engines A and B are idle.

Pricing: funding grid to option budget

USD funding interpolated to 18M and swapped into KRW

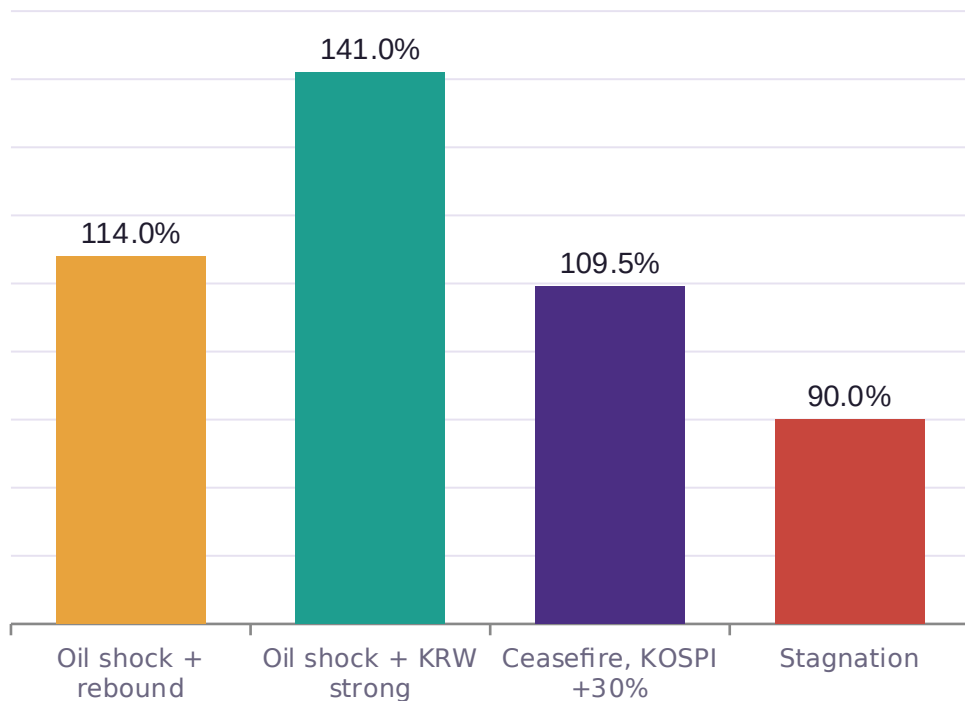


Protection dial: the fund chooses its risk

Protection	Budget	A participation	B payout	C participation	Max loss
90% (recommended)	~14%	~120%	~27%	~65%	10% (~KRW 13.5bn)
80% (aggressive)	~24%	~220%	~45%	~110%	20%

Wins in war and peace, not stagnation

Note redemption at 18 months, 90% protection case



- Base: oil shock, KOSPI +20% from trigger**
A pays 24%
- Best: oil shock, KRW resilient, rebound**
A + B pay 51%
- Alternative: ceasefire and re-rating**
C pays 19.5%
- Worst: flat, or no rebound after shock**
Capped at -10% at maturity

Illustrative outcomes using budget split on slide 3. Redemption = 90% + engine payouts.

Sizing and market risk

Unit 15 sizing identity, VaR / ES, and stress tests with spot and vol shocked together

Sizing identity

Risk budget: 25 bp of USD 10bn AUM

= **USD 25M**

Loss per unit: 10% of notional
(contractual, held to maturity)

Max notional = $25\text{M} \div 10\%$
= **USD 250M**

Proposed KRW 135bn $\approx$ USD 97M
uses 39% of the budget. Max loss
 $\approx$ USD 9.7M = 10 bp of AUM, a
third of the USD 30M lost with the
previous manager.

VaR and Greeks (% notional)

Fair value at issue	98.1%
10-day 99% VaR (delta-normal)	5.6%
Expected shortfall 99%	6.4%
Delta Brent / +1%	+0.33
Delta KOSPI 200 / +1%	+0.26
Delta USD/KRW / +1%	-0.30
Vega KOSPI / +1 vol pt	+0.36
Worst MTM before maturity	~84%

Instant stress: MTM change

Risk-off **-0.3 pts**

KOSPI -20%, KRW -5%, vols +10 pts,
rates +50 bp

Oil shock **+13.0 pts**

Brent +30%, KOSPI -10%, vols +8 pts

Ceasefire **-2.9 pts**

Brent -20%, KOSPI +10%, vols -3 pts

*Long vega cushions the risk-off case:
higher vol lifts Engine A.*

Risk register and exit plan

Every risk named, and four exits defined in advance

Risk	Assessment and mitigant
Model	Correlation and skew drive B and C. Price with surface vols; show correlation sensitivity
Liquidity	OTC note: ~1.9% margin in, ~1% bid/offer out. Hedges trade in deep KOSPI 200, Brent and USD/KRW markets
Counterparty	Natixis issuer risk: disclose rating and CDS. ISDA/CSA for the OTC hedges
Operational	Daily Brent trigger monitoring, independent fixing source, monthly MTM report
KRW mechanics	CCS basis and offshore non-deliverability: settlement terms agreed up front

1

Profit target

Sell back to Natixis when MTM $\geq$ 115% (net of ~1% bid/offer)

2

Thesis invalidation

Exit if WGBI flows reverse and MSCI drops Korea from review, even if P&L is positive

3

Time stop

Month 12: Engine A's trigger window closes. Hold or sell back on B and C value

4

No stop-loss as risk control

Max loss is contractual (Unit 15). Any stop triggers on Brent/KOSPI levels, not note price

Appendix: terms and methodology

Not counted toward the 10-slide limit

Issuer	Natixis
Notional	KRW 135bn
Trade / issue date	17 Sep 2026
Maturity	18 months
Capital protection	90% at maturity
Underlyings	Brent (CO1 Comdty), KOSPI 200 (KOSPI2 Index), USD/KRW (KRW Curncy)
Funding	USD grid 18M 5.04%, swapped to KRW via CCS

Methodology

- Monte Carlo, 200k paths, daily steps, correlated GBM for Brent, KOSPI 200 and USD/KRW
- Placeholder inputs: vols 35% / 22% / 9%; KRW 2.5%, USD 3.75%; KOSPI dividend 1.8%
- Engine C priced with skew (130% strike at lower IV)
- To finalise: Bloomberg OVDV surfaces, 25-delta risk reversals, realised correlations, CCS basis, local-vol pricing for barriers
- Considered and rejected: Korea Value-Up Index as Engine C underlying. Hedge liquidity is far thinner than KOSPI 200 (evidence OI and volume as of 17 Sep 2026)
- Next steps: agree protection level and weights with the PM; indicative term sheet in 48 hours; final pricing on trade date